CPSC 481 Project 2: Bayesian Network Diagnosis with Gemini

(a) Why this LLM

We chose Gemini 3.8 Flash (through Google's `google-genai` SDK) because it is Google's current stable Flash model and supports both structured JSON output and an adjustable "thinking" level for multi-step arithmetic. The assignment's suggested Gemini 2.5 Flash is now limited to accounts that have used it before.

(b) Major challenge

The main challenge was making the model's answer both correct and machine-readable. Exact inference means summing over every combination of the unobserved variables, and a free-text reply would be hard to parse. We gave the model the full network, the observed evidence and the enumeration method, set its thinking level to "high", and required a JSON reply matching a schema in which the disease must be TB, Cancer or Bronchitis; we validate it and retry up to twice. API churn added to this: Google's docs show two different structured-output APIs, so we confirmed the call shape against the installed SDK before writing any code.

(c) Something surprising

With thinking on, Gemini matched exact enumeration to four decimal places on every case we tested (the spec case and an 11-row sample), even when the top two diseases were only 0.0064 apart. However, each answer took about 40-55 seconds, where `enumeration_ask` returns almost instantly. Google also advises against lowering the temperature for Gemini 3 models, which is the opposite of the usual "use a low temperature for math" advice.